

Activity 04: “Design the University Tech Hub”

This is a team activity. You are building one design together, and everyone will receive the same score for the final product. However, everyone is expected to participate in the design discussion.

Overall Problem Statement:

California Lutheran University is creating a new Technology & Innovation Hub.

The hub contains several Labs. Each Lab contains a number of Workstations. A workstation is specifically installed in and managed as part of its lab.

Students may use different labs during the semester. A student can use many labs, and each lab can be used by many students.

Each lab has a Faculty Supervisor. A faculty member may supervise more than one lab, and faculty members continue to exist independently of any particular lab.

The university also owns a collection of Robots that can be assigned to labs for projects. A robot may be moved from one lab to another.

Each workstation contains a CPU and Memory modules that are modeled as integral parts of that workstation.

⇒ Your team has been asked to design an object-oriented model for this system before anyone starts writing Java code.

Phase 1 — What are our objects? [5 mins, at least]

⇒ Probable list:

- TechnologyHub
- Lab
- Workstation
- Student
- FacultySupervisor
- Robot
- ...
- ...
- ...

Phase 2 — Relationship Detective [10 mins]

Identify the relationships and the multiplicity between all the relevant parties.

Example:

A Student can use several Labs. A Lab can be used by many Students.

Student * ----- * Lab

Prepare a table with the following columns and enlist the relationship in appropriate column:

Class A	Relationship	Class B	Multiplicity	Why?
Student	Association	Lab	* : *	Students and labs exist independently
...	...	...	...	...
...	...	...	...	...

Phase 3 — Build ONE UML Diagram [15~20min]

- ➔ On paper/ one common shared file
- ➔ Online Resources [Preferred]:
 - <https://mermaid.js.org/>
 - Draw.io (Diagrams.net)

Submit the UML Diagram or the link to your diagram and the responses you made for phase 1 and 2.

Tentative Grading Rubric:

Component	Points	What you're looking for
Participation & Teamwork	4	Contributes ideas, discusses decisions, helps the group
Class Identification	3	Reasonable classes selected from the requirements
Relationships	5	Association/Aggregation/Composition identified appropriately and justified
Multiplicity	3	Correct multiplicities on relationships
Final UML Diagram	5	Complete, readable, consistent with team's analysis
Total	20	